Yashovardhan Mukerji

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.71
Sir Padampat Singhania Education Centre	Class XII (CBSE)	2023	94.40%
Sir Padampat Singhania Education Centre	Class X (CBSE)	2021	99.60%

EXPERIENCE

Quant Intern, Futures First

- Co-built a Rate Monitor and RBA Dashboard tracking yields, inflation, labour, and growth indicators; developed an NLP-based n-gram tool to parse RBA communications and derive thematic sentiment on policy tone.

- Built statistical and pricing tools - z-score outlier detectors, OHLCV visualizers, implied OIS rate-curve and meeting-premia calculators, and fair-value models - for IB/IR futures contracts.

- Designed and backtested systematic strategies (MA crossover, mean-reversion fly) and a linear regression model (R2 = 88%) linking IR and IB contract pricing.

- Built scenario-based (Hawkish/Dovish/In-Line) rate curves around RBA and Fed policy events, translating CPI and Labour Force releases into structured trades with tracked PnL.

SKILLS

- Languages : Python, C++, C, SQL, Bloomberg Query Language, CUDA
- Data Science and AI/ML : Regression, Clustering, Z-Score analysis, Bagging, Boosting, Nueral Network modeling.
- Softwares/Applications : Bloomberg Terminal, Excel, Jupyter, VS code, Scikit, Antigravity

PROJECTS

Multi-Objective Optimization of Airline Profit vs. Cost (ANN + NSGA-II)

- Selected 9 continuous inputs and 2 conflicting outputs (profit, operating cost) from a 7,974-sample airline route dataset.
- Trained and validated an ANN surrogate model (Adam optimizer, MinMax-scaled data, 70/15/15 split).
- Achieved R^2 0.93 with low RMSE across train, validation, and test sets, with no overfitting.
- Coupled the ANN with NSGA-II (pymoo, population 100, 100 generations) to search the input space.
- Generated a Pareto front of non-dominated profit-cost trade-off solutions.

A Simple Plane-Wave DFT Code for Cubic Solids

- Studied a plane-wave Density Functional Theory implementation built on the DFT++ operator formalism
- Benchmarked total energies for H, He, and H to within 0.02% of reference values.
- Visualized electron densities, including covalent bonding in H.
- Independently implemented a 2D plane-wave DFT code from scratch in NumPy, covering the kinetic-energy operator, a Fourier-space Poisson solver, and a Rayleigh-quotient SCF loop with oscillation detection.
- Compared the design choices of the 2D code against the reference 3D implementation.

Molecular Dynamics Simulation of Argon Melting

- Built an MD engine from scratch in Python to model bulk Argon using the Lennard-Jones potential.
- Initialized atoms on an FCC lattice with periodic boundary conditions and the minimum-image convention.
- Implemented velocity-Verlet time integration with a Verlet neighbor list for efficient force evaluation.
- Ran a two-phase simulation (velocity-rescaling equilibration, then production) to sample thermodynamic properties.

- Verified energy conservation and computed the radial distribution function and observed melting across target temperatures and densities.

Modelling Poisson's, Heat, and Laplace's Equations

- Solved Poisson's equation in 1D and 2D using finite-difference discretization and Jacobi iterative relaxation.
- Validated convergence against analytical solutions.
- Extended the framework to Laplace's equation, and to the Heat and Poisson equations using FFT-based spectral methods.
- Compared accuracy and runtime of the spectral approach against the iterative approach.
- Implemented in Python (NumPy, SciPy, Matplotlib) with visualizations of convergence behavior and solution fields.

POSITION OF RESPONSIBILITY

Head of Public Relations & Networking, E-Cell IIT Hyderabad

- Led E-Merge, Start-Up Fair, and Founders Hive, driving nationwide outreach and connecting startups with VCs, investors, and CXOs across multiple cities.
- Spearheaded an experiential entrepreneurship program for school students and streamlined PR workflows to accelerate execution of key initiatives.
- Organized Founders Hive Bangalore, bringing together 10 early-stage startups and 2 seasoned mentors for product showcases and expert feedback.

Manager, Public Relations & Networking, E-Cell IIT Hyderabad

Led execution of the Start-Up Fair (500+ attendees) and E-Merge initiatives engaging 200+ students; secured sponsor-

ships and mentored five students.

EXTRACURRICULAR

Powerlifting - Guitar